

FREEDOM INTERNATIONAL SCHOOL
MATHEMATICS
TOPIC: SUCCESSIVE DIFFERENTIATION

CLASS : XII

NAME :

1. If $x = a \cos \theta + b \sin \theta$, $y = a \sin \theta - b \cos \theta$, show that

$$y^2 \frac{d^2 y}{d x^2} - x \frac{d y}{d x} + y = 0.$$

2. $y = x^x$, prove that $\frac{d^2 y}{d x^2} - \frac{1}{y} \left(\frac{d y}{d x} \right)^2 - \frac{y}{x} = 0$

3. If $y = e^{m \sin^{-1} x}$, then show that $(1-x^2) \frac{d^2 y}{d x^2} - x \frac{d y}{d x} - m^2 y = 0$.

4. If $y = e^{\tan^{-1} x}$, then show that $(1+x^2) \frac{d^2 y}{d x^2} + (2x - 1) \frac{d y}{d x} = 0$.

5. If $y = e^{a \cos^{-1} x}$, $-1 \leq x \leq 1$, show that $(1 - x^2) \frac{d^2 y}{d x^2} - x \frac{d y}{d x} - a^2 y = 0$.

6. If $y = e^{ax} \cos bx$, show that $\frac{d^2 y}{d x^2} - 2a \frac{d y}{d x} + (a^2 + b^2)y = 0$.

7. If $y = e^x \sin x$, show that $\frac{d^2 y}{d x^2} - 2 \frac{d y}{d x} + 2y = 0$.

8. If $y = \csc^{-1} x$, $x > 1$, then show that $x(x^2-1) \frac{d^2 y}{d x^2} - (2x^2-1) \frac{d y}{d x} = 0$.

9. If $y = \frac{\sin^{-1} x}{\sqrt{1-x^2}}$, show that $(1-x^2) \frac{d^2 y}{d x^2} - 3x \frac{d y}{d x} - y = 0$.

10. If $y = \tan^{-1} x$, show that $(1+x^2) \frac{d^2 y}{d x^2} + 2x \frac{d y}{d x} = 0$.

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1. If $y = (\cot^{-1} x)^2$, show that $(1 + x^2)^2 \frac{d^2y}{dx^2} + 2x(1+x^2) \frac{dy}{dx} = 2$.
2. If $y = (\tan^{-1} x)^2$, show that $(1 + x^2)^2 \frac{d^2y}{dx^2} + 2x(1+x^2) \frac{dy}{dx} = 2$.
3. If $y = \{x + \sqrt{1 + x^2}\}^m$, then show that $(1+x^2) \frac{d^2y}{dx^2} + (x) \frac{dy}{dx} - m^2y = 0$.
4. If $y = \log[x + \sqrt{a^2 + x^2}]$, then show that $(a^2+x^2) \frac{d^2y}{dx^2} + x \frac{dy}{dx} = 0$.
5. If $y = (\log[x + \sqrt{1 + x^2}])^2$, then show that $(1+x^2) \frac{d^2y}{dx^2} + x \frac{dy}{dx} - 2 = 0$.
6. If $y = 3\cos(\log x) + 4\sin(\log x)$ then show that $x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx} + y = 0$.
7. If $y = \sin(\sin x)$, prove that $\frac{d^2y}{dx^2} + (\tan x) \frac{dy}{dx} + y \cos^2 x = 0$.
8. If $y = A \cos(nx) + B \sin(nx)$, prove that $\frac{d^2y}{dx^2} + n^2y = 0$.
9. If $y = e^x(\sin x + \cos x)$, prove that $\frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + 2y = 0$.
10. If $e^y(x+1) = 1$, prove that $\frac{d^2y}{dx^2} = \left(\frac{dy}{dx}\right)^2$.

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Differentiate each of the following w.r.t. x:

1. $y \sec x + \tan x + x^2 y = 0$

2. $y \tan x - y^2 \cos x + 2x = 0$

3. $\cot(xy) + xy = y$

4. $\tan(x+y) + \tan(x-y) = 1$

5. $xy \log(x+y) = \tan^{-1} \left(\frac{y}{x} \right)$

6. $e^x \log y = \sin^{-1} x + \sin^{-1} y$

7. $x^2 + y^2 = \log(xy)$

8. $10. \sin(xy) + \frac{x}{y} = x^2 y$

9. $\sin(x+y) = x^3 + y^3$

10. $3\sin(xy) + 4\cos(xy) = 5$

11. $x^4 + y^4 - a^2 xy = 0$

12. $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

13. If $\cos(xy) = x$, show that $\frac{dy}{dx} + \frac{1 + y \sin(xy)}{x \sin(xy)} = 0$.

14. $y = x \sin y$, then show that $\frac{dy}{dx} + \frac{\sin y}{1 - x \cos y} = 0$

15. If $\cos^{-1} \left(\frac{x^2 - y^2}{x^2 + y^2} \right) = 2a$, show that $y \frac{dy}{dx} = x \tan^2 a$

16. If $\tan^{-1} \left(\frac{x^2 - y^2}{x^2 + y^2} \right) = a$, show that $\frac{dy}{dx} = \frac{x(1 - \tan a)}{y(1 + \tan a)}$

17. If $e^x + e^y = e^{x+y}$, prove that $\frac{dy}{dx} = -e^{y-x}$

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1. If $\sin y = x \sin(a+y)$, prove that $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$
2. If $\cos y = x \cos(a+y)$, prove that $\frac{dy}{dx} = \frac{\cos^2(a+y)}{\sin a}$
3. If $x \sin(a+y) + \sin a \cos(a+y) = 0$, prove that $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin a}$
4. If $y = x \sin(a+y)$, PT $\frac{dy}{dx} = \frac{\sin^2(a+y)}{\sin(a+y) - y \cos(a+y)}$
5. If $\sin y = x \cos(a+y)$, prove that $\frac{dy}{dx} = \frac{\cos^2(a+y)}{\cos a}$
6. If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$, then PT $\frac{dy}{dx} = \frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$
7. If $x\sqrt{1+y} + y\sqrt{1+x} = 0$, PT $(1+x)^2 \frac{dy}{dx} + 1 = 0$
8. If $y\sqrt{1+x^2} = \log(\sqrt{1+x^2} - x)$, PT $(1+x^2) \frac{dy}{dx} + xy + 1 = 0$
9. If $y\sqrt{1-x^2} + x\sqrt{1-y^2} = 1$, PT $\frac{dy}{dx} = -\frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}$
10. If $y = x^x + x^{1/x}$, find $\frac{dy}{dx}$
11. If $x^y \cdot y^x = 1$, find $\frac{dy}{dx}$
12. If $x^y + y^x = (x+y)^{x+y}$, find $\frac{dy}{dx}$
13. If $x^m \cdot y^n = 1$, ST $\frac{dy}{dx} = -\frac{my}{nx}$
14. If $x^{13} y^7 = (x+y)^{20}$, PT $\frac{dy}{dx} = \frac{y}{x}$
15. If $y = \sin(x^x)$, PT $\frac{dy}{dx} = \cos(x^x) \cdot x^x(1+\log x)$
16. If $y = \frac{(x^2-1)^3(2x-1)}{\sqrt{(x-3)(4x-1)}}$, then find $\frac{dy}{dx}$

**FREEDOM INTERNATIONAL SCHOOL
MATHEMATICS**

TOPIC: DIFFERENTIATION FOR INFINITE SERIES

CLASS : XII

NAME :

1. If $y = \sqrt{x}^{\sqrt{x}^{\sqrt{x}^{\dots\infty}}}$, ST $\frac{dy}{dx} = \frac{y^2}{x(2-y\log x)}$

2. If $y = (\cos x)^{(\cos x)^{(\cos x)^{\dots\infty}}}$, ST $\frac{dy}{dx} = \frac{y^2 \tan x}{(1-y\log \cos x)}$

3. If $y = (\tan x)^{(\tan x)^{(\tan x)^{\dots\infty}}}$, ST $\frac{dy}{dx} = 2$ at $x = \frac{\pi}{4}$

4. If $y = e^{x+e^{x+e^{x+\dots\infty}}}$, ST $\frac{dy}{dx} = \frac{y}{1-y}$

5. If $y = (\sin x)^{(\sin x)^{(\sin x)^{\dots\infty}}}$, ST $\frac{dy}{dx} = \frac{y^2 \cot x}{(1-y\log \sin x)}$

6. If $y = \sqrt{\cos x + \sqrt{\cos x + \sqrt{\cos x + \dots \text{to } \infty}}}$, PT $\frac{dy}{dx} = \frac{\sin x}{1-2y}$

7. If $y = \sqrt{\sin x + \sqrt{\sin x + \sqrt{\sin x + \dots \text{to } \infty}}}$, PT $\frac{dy}{dx} = \frac{\cos x}{2y-1}$

8. If $y = x + \frac{1}{x + \frac{1}{x + \frac{1}{x + \dots}}}$, PT $\frac{dy}{dx} = \frac{y}{2y-x}$

9. If $y = \sqrt{\log x + \sqrt{\log x + \sqrt{\log x + \dots \text{to } \infty}}}$, PT $x \frac{dy}{dx} = \frac{1}{2y-1}$

10. If $y = \sqrt{\tan x + \sqrt{\tan x + \sqrt{\tan x + \dots \text{to } \infty}}}$, PT $\frac{dy}{dx} = \frac{\sec^2 x}{2y-1}$